

Computational Optics and Biomedical Imaging Simulations

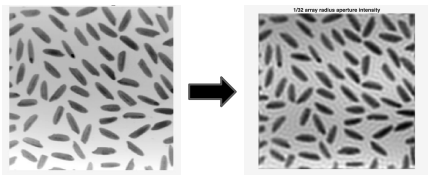
[Project Source](#)

MATLAB Simulations | Fourier Optics | Phase Imaging | Microscopy | OCT

Phase Contrast Microscopy

Zernike Phase Contrast

Original Phase Object (Rice) Zernike Phase Contrast
(Phase Dot 1/32 Array)

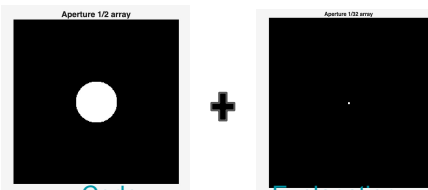


Simulated phase imaging of samples using Fourier optics and Zernike phase contrast techniques. The phase dot converts phase shifts into visible intensity variation.

Aperture (Fourier Plane)

Main Aperture
($\frac{1}{2}$ array radius)

Phase Dot
($\frac{1}{32}$ array diameter)

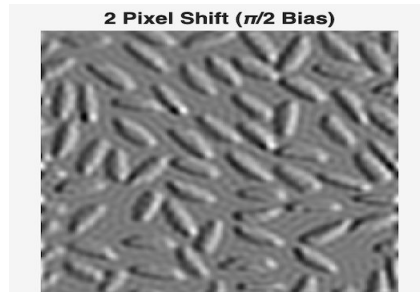


[Code](#)

[Explanation](#)

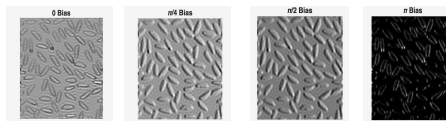
Differential Interference Contrast (DIC)

DIC Result (Shear ± 2 px, $\pi/2$ Bias)



Implemented DIC by laterally shearing the image field and applying a phase bias. $\pi/2$ bias produces maximum contrast, revealing optical path differences as pseudo-3D shadows.

Bias Comparison (shear ± 2 px)

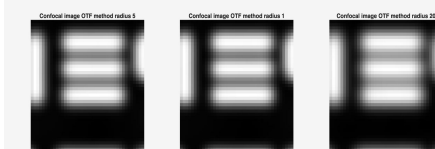


Increasing Phase Bias

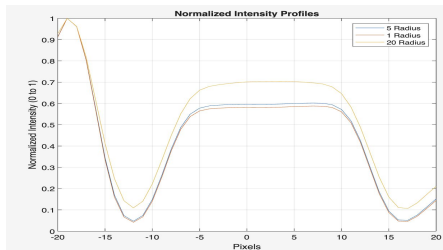
Confocal Microscopy (OTF Method)

Confocal Images - Different Pinhole Radii

Radius = 1 Radius = 5 Radius = 20



Normalized Intensity Profiles



Simulated laser-scanning confocal microscopy using the OTF method. Similar pinholes reject more out-of-focus light, yielding higher resolution and better optical sectioning

Best Resolution: Radius = 1 pixel

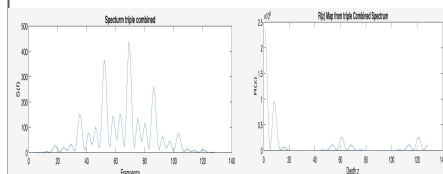
[Code](#)

[Explanation](#)

Optical Coherence Tomography (OCT Method)

Spectrum $|S(f)|^2$
(Combined Pulse)

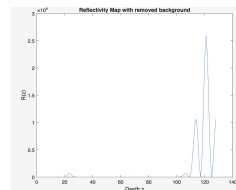
Reflectivity Map $R(z)$



Peaks at: $z * 8$ (mirror), 61 (interference of 180 & 120), 121 (reference & 120). (180 wraps around due to 128 points.)

Modeled OCT using time-delayed Gaussian pulses. Interference in the spectrum generates a depth-resolved reflectivity map revealing sample structures.

Reflectivity Map with Background Removed



- Strong reference dominates near $z = 0$
- Removing first 20 points removes interference and reveals sample peaks

[Code](#)

[Explanation](#)